

Cesar Briones

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Education

University of South Florida – M.S. in Aerospace Engineering	Expected May 2028
University of South Florida – B.S. in Mechanical Engineering	May 2026
Key Courses: Aerodynamics, Propulsion, Vibrations, Finite Element Methods	GPA: 3.72/4.0

Skills

CAD/Coding:	Siemens NX, SolidWorks, Python, MATLAB, Simulink, C#, LaTeX, Epsilon3
Design:	GD&T, DFMA, P&ID, Prototyping, CNC Milling Programming and Operation, Soldering
Analysis Tools:	ANSYS Workbench / Mechanical / Fluent, Excel, LabVIEW
Certifications:	CSWP Simulation, CSWA Design, CSWA Additive Manufacturing

Projects

Rocket Engine Development	July 2025 – Present
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- Designed a N₂O/ethanol 1.6kN liquid engine performing thermochemical analysis and nozzle optimization for 0.9 atm
- Conducted transient thermal analysis to size copper combustion chamber walls as a heat sink, calculating thermal penetration depth and temperature gradients for target burn duration of ~7 seconds.
- Performed CFD in ANSYS Fluent to extract discharge coefficients from mass flow predictions and spray cone angle
- Produced GD&T drawings for all CNC-machined components, outsourcing fabrication to third-party vendors
- Iterated on P&ID documentation for propellant feed system, refining instrumentation layout across design revisions

Experience

Society of Aeronautics and Rocketry (USF Rocketry Team)	Tampa, FL
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<i>Chief Engineer – IREC 10K Student Competition</i>	May 2025 - Present
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- Responsible for technical development, systems engineering and integration of 7 subsystems in student-built rocket
- Supervised all manufacturing operations of 25+ aluminum components via CNC programming, vertical mill operations, drill press work, and tapping, ensuring dimensional accuracy, resulting in the year with most in house made parts
- Conducted system design reviews to confirm readiness, coordinating with faculty mentors and competition judges
- Established top-level mission requirements and design constraints through trade studies and competition rule analysis
- Mentored 12+ junior members in CAD/CAM, CNC machining, FEA, CFD, and systems engineering principles

<i>Airbrakes Lead – NASA Student Launch Competition</i>	Jun 2024 – May 2025
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- Executed parametric CFD simulations in ANSYS Fluent across deployment angles (0-100%) and Mach regimes (0.3 - 0.7), characterizing drag force profiles to support flight control algorithm development geometry and torque
- Designed a four-bar linkage mechanism for drag control, performing kinematic analysis to optimize deployment
- Developed a regression model in MATLAB from CFD, enabling fast drag lookup for the PID control system
- Validated CFD-predicted force through experimental flight testing, comparing data against simulation results
- Achieved 26% mass reduction in CNC-machined aluminum components through iterative FEA-driven design optimization while maintaining structural safety factors, allowing to stay within the team derived mass requirement.

USF Corrosion Research Laboratory	Tampa, FL
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<i>Instrumentation Researcher – Corrosion prediction in tendons</i>	May 2024 – Dec 2025
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- Designed a 2-layer PCB using EasyEDA and SMD components, reducing assembly complexity and cost replication
- Developed FFT-based signal processing algorithms in Python to detect voids in post-tensioned tendons by comparing measured impedance against reconstructed baseline signatures, implementing filtering and polynomial correction
- Developed a real-time wireless data acquisition pipeline with Arduino-to-PC serial communication (UART) at 115200 baud, processing extracted sensor data in Python with automated anomaly detection and live visualization
- Designed 3D-printed enclosures and sensor mounting hardware for portable field deployment, enabling rapid prototyping iterations and tool-free assembly for bridge structure inspections

RANCS Research Laboratory	Tampa, FL
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<i>Research Lab Assistant – Autonomous Systems</i>	Dec 2023 – Mar 2024
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- Designed and built a t-slot aluminum frame to elevate a \$10,000 LIDAR sensor, improving field of view and eliminating interferences with vehicle roof, enabling 100+ hours of successful autonomous vehicle testing
- Coordinated with PI and graduate students to define LIDAR mounting requirements, designing a cost-effective solution using existing lab inventory of t-slot aluminum extrusions and hardware